Root Cause Analysis (RCA) Report

Module 4: Problem Solving & Root Cause Analysis

1. Incident Summary

Users reported that they were unable to log in to the InstaSafe Portal. Although the login page was accessible, authentication requests were failing, preventing users from accessing business applications. The issue affected multiple users and interrupted normal business operations until the authentication service was restored.

2. Problem Statement

Issue: Users were unable to authenticate successfully to the InstaSafe Portal.

Symptoms

- Login page loaded successfully.
- Username and password were accepted.
- Authentication failed after login attempt.
- Users could not access protected applications.
- Increase in user complaints and support tickets.

3. Impact Analysis

Business Impact

- Remote users could not access company resources.
- Employee productivity was reduced.
- Business operations experienced temporary disruption.

Technical Impact

- Authentication service failed to validate user login requests.
- SAML authentication process was interrupted.
- Access to protected applications was unavailable.

4. Investigation Performed

The following troubleshooting steps were carried out to identify the cause of the issue:

- Verified user credentials.
- Confirmed InstaSafe Portal availability.
- Reviewed authentication logs.
- Checked Identity Provider (IdP) connectivity.
- Verified SAML authentication configuration.
- Examined recent configuration changes.
- Compared current configuration with the previous working version.

5. Whys Analysis

Step	Why	Answer
1	Why were users unable to log in?	Authentication requests were failing.
2	Why were authentication requests failing?	The Identity Provider was not validating SAML requests.
3	Why was the Identity Provider not validating requests?	The SAML metadata configuration was incorrect.

Step	Why	Answer
4	Why was the SAML metadata incorrect?	Incorrect configuration was deployed during a recent update.
5	Why was incorrect configuration deployed?	There was no validation or testing before deployment.

6. Root Cause

The root cause of the incident was the deployment of incorrect **SAML metadata configuration** without proper validation or testing. As a result, the Identity Provider could not successfully authenticate users, causing all login requests to fail.

7. Corrective Actions Taken

The following actions were taken to resolve the incident:

- Restored the previous working SAML metadata configuration.
- Restarted the authentication service.
- Verified communication between the Identity Provider and InstaSafe Portal.
- Performed successful login testing using test accounts.
- Monitored authentication logs to confirm the issue was resolved.

8. Preventive Actions

To prevent similar incidents in the future, the following recommendations are proposed:

- Validate all SAML configuration changes before deployment.
- Implement peer review for authentication-related configuration updates.
- Test configuration changes in a staging environment before production deployment.

- Enable automated validation for SAML metadata.
- Maintain configuration backups before every deployment.
- Implement monitoring and alerting for authentication failures.
- Establish a documented rollback procedure for failed deployments.
- Update change management documentation for authentication services.

9. Lessons Learned

- Authentication configuration changes should always be validated before deployment.
- Structured troubleshooting techniques such as the 5 Whys help identify the true root cause rather than only addressing symptoms.
- Testing in a staging environment significantly reduces production risks.
- Continuous monitoring enables faster detection and resolution of authentication issues.

10. Conclusion

The login issue was successfully investigated using the 5 Whys technique and Root Cause Analysis methodology. The investigation identified an incorrect SAML metadata configuration as the root cause. Corrective actions restored authentication services, and preventive measures have been recommended to reduce the likelihood of similar incidents in the future.